Experimental Demonstration of the Quantum Zeno Effect

Nova Royle[1]

[1] *Department of Physics and Astronomy, University of California, Los Angeles, CA 90095, U.S.A.*

(Dated: June 12, 2026)

The quantum Zeno effect (QZE) predicts that the evolution of a quantum state can be influenced by the presence of external measurements of the state. Here, we experimentally verify the presence of the QZE by manipulating the polarization of coincident photons. Fair agreement to the theoretical fit of $R^2 = 98.1\,\%$ was observed. We also see an increase in the coincidence count rate with additional polarizer stages with significance $5.80\,\sigma$, providing strong evidence for the QZE.

Introduction—The quantum Zeno effect (QZE) was first theoretically predicted by Misra and Sudarshan in 1977 while investigating probabilities for state decays [1]. Repeated measurements of a quantum state cause repeated wavefunction collapse into the measured state, so it cannot evolve from that state.[1] The effect most often applies to time-dependent state evolution, but the principle can be expanded into time-independent regimes [2].

One such regime is the case of optical polarization, where standard predictions predict no transmission between horizontally and vertically polarized light. However, with the inclusion of optical polarizers, the QZE predict the photon wavefunctions will successively collapse into the polarizer direction. With infinite polarizers, essentially all photons will be transmitted despite the input and output polarization being 90° apart. The result of the QZE is often used to optimize interaction-free measurements [3, 4]. Here, we show the presence of the QZE through observing coincident photons through a varying number of optical polarizers. The expected behavior due to the QZE will be derived theoretically and verified with experimental data.

Theory—The Zeno effect will be demonstrated here by considering the evolution between polarization states, where the basis states are $|H\rangle$ and $|V\rangle$. An arbitrary linear polarization state can then be written as

$$|\psi\rangle = \cos(\theta)\,|H\rangle + \sin\theta\,|V\rangle\,, \tag{1}$$

where θ is the angle of polarization relative to $|H\rangle$.

We then aim to calculate the magnitude of the state that transmits through the polarizer. Allowing the photon to interact with the polarizer is effectively "measuring" the state of the photon, which causes its polarization wavefunction to collapse into the direction of the polarizer. In other words, we want to calculate the magnitude of state projected onto the polarizer axis. Without loss of generality, we will assume the polarizer is set to the $|H\rangle$ axis. If the polarizer is set to a different angle α, this is equivalent to simply rotating the polarization state by $-\alpha$. The projection operator onto the $|H\rangle$ state is given by [5],

$$P_H = |H\rangle\,\langle H|\,.$$

Thus, projecting the arbitrary linear polarization onto $|H\rangle$ gives,

$$\begin{aligned} P_H\,|\psi\rangle &= |H\rangle\,\langle H|\,(\cos(\theta)\,|H\rangle + \sin\theta\,|V\rangle) \\ &= \cos(\theta)\,|H\rangle\,. \end{aligned}$$

Notice that this state no longer has any component in the $|V\rangle$ direction. The magnitude of this state gives the probability that a given photon is transmitted through the polarizer (and thus has a polarization aligned to the detector):

$$\langle\psi|\psi\rangle = \cos^2(\theta),$$

If there is a stream of photons entering the polarizer, the rate of output photons is proportional to the probability of transmission,

$$\langle\psi|\psi\rangle \propto \Gamma,$$

where Γ is the coincidence count rate. Thus, the relation between the input and output count rate is given by

$$\Gamma(\theta) = \Gamma_{\max}\cos^2(\theta). \tag{2}$$

For classical theories of light as an electromagnetic wave, this relation considering $\Gamma \leftrightarrow I$ is also known as Malus's law [6].

In this experiment, we wish to verify this relation is true for single photons rather than electromagnetic waves. We study the photon as it traverses multiple polarizers, each collapsing the photon's wavefunction into its polarization direction. If there are n polarizers set so that the angle θ between them is consistent, the photon is experiencing successive measurements in different measurement bases (polarization states). Thus, the coincidence count rate for n polarizers with angle difference θ is

$$\Gamma(\theta, n) = \Gamma_{\max}\cos^{2n}(\theta).$$

[1] One might refer to the effect as "a watched state never boils."

In the case where the total angle difference is 90°, the relation between n and θ is

$$\theta = \frac{90^\circ}{n} = \frac{\pi}{2n}. \tag{3}$$

This leads to expected expression for the coincidence count rate for the experiment [3, 7],

$$\Gamma(n) = \Gamma_{\max} \cos^{2n}\left(\frac{\pi}{2n}\right). \tag{4}$$

It is critical to note that for increasing n, the rate of coincident photons also increases. Interestingly, for $n \to \infty$, $\Gamma \to \Gamma_{\max}$, meaning that all photons pass through despite the initial and final polarization angles being 90° apart. Thus, the QZE predicts that as more polarizers are added, more of the photons will transmit through, as the more frequent observations force wavefunction collapse in the polarizer bases. Without the QZE, a zero slope between Γ and n should be observed.

Experimental Methods—To ensure measurements are characteristic of the QZE, coincident photons must be created and measured. The process of spontaneous parametric down-conversion (SPDC) was used to generate two photons with wavelength 810 nm from a single pump photon with wavelength 405 nm [8–10]. We use beta-barium borate, which causes type I SPDC which ensures the down-converted photons are in the same polarization state [11]. Thus, these two output photons (sent to arm 0 and arm 1) are present in the state

$$|\psi\rangle = |H\rangle_0 |H\rangle_1 .$$

In this case, the polarizations of the arm 0 and arm 1 states match exactly and have no $|V\rangle$ component. We have not created an entangled state, but such a state is not necessary to reveal the QZE. The SPDC efficiency is $\approx 10^{-11}$, so few photons are down-converted. The rest are blocked by a filter and do not continue through the optical setup.

Using coupling mirrors, the photons are redirected towards fiber couplers attached to two inputs on a photon counter, shown in Fig. 1. In arm 0, between the mirror and fiber coupler are four identical polarizers on rotation mounts, which are arbitrarily adjustable and nominally set to the vertical direction parallel to $|V\rangle$. Ensuring the number of physical polarizers present is constant helps to remove the confounding variable of a nonconstant transmission coefficient. The photons in arm 1 are simply coupled into an optical fiber directly as a control.

The photon counter is able to detect the single count rates and coincidence count rates, which is used in order to reduce the effects of background light reaching the coupler. Simultaneous clicks in arm 0 and arm 1 within the coincidence window are recorded as a coincident photon.

Experiments were performed with the quED Entanglement Demonstrator produced by qutools GmbH [10].

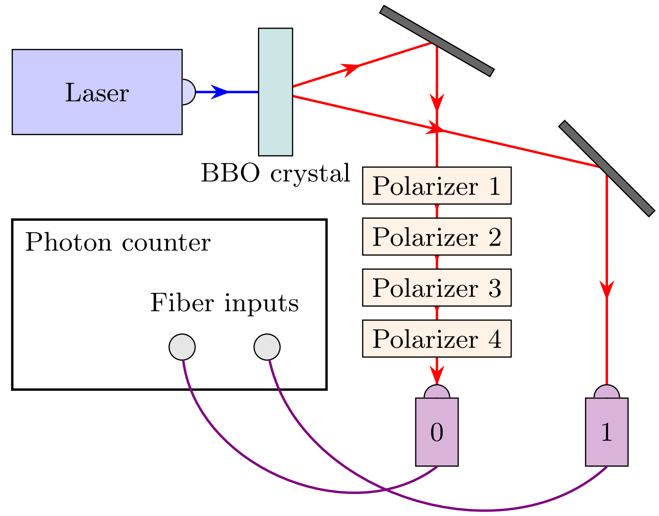

FIG. 1. Experimental setup. The laser diode outputs 405 nm horizontally-polarized light, which passes through a nonlinear down-conversion crystal (beta-barium borate, or BBO), which causes the incident photons to undergo spontaneous parametric down-conversion [11]. This then generates pairs of horizontally-polarized 810 nm photons, which then are separated into two arms. The photon in arm 0 passes through 4 successive polarizers, whereas the photon in arm 1 passes through free space. At the end of each arm, the photons are coupled into optical fibers and led into the input of a photon counter that detects coincident photons.

The photon counter is the qutools GmbH quCR Control and Read-out Unit [12], provided with the quED apparatus.

To begin the experiment, the 405 nm wavelength laser diode was set to its operating current of 38 mA. Beam alignment was performed to optimize single count rates and coincident count rates by adjusting the coupling mirror and fiber collimator. Once the alignment of the laser was optimized, the QZE experiment could begin. To increase the number of relevant polarizers, the first n polarizers are set at angles according to Equation 3. The count rates were then tabulated and analyzed.

Results—This experiment was performed for up to 4 polarizers, set at equal angles from 0° to 90°. Thus, $\Delta\theta$ ranged from 90° to 22.5°. Error analysis was performed assuming a Poisson distribution for the number of counts,

$$\sigma = \sqrt{N},$$

and an error of 0.5° in the angles, which were marked in 1° increments on the rotation mount. Error propagation for all following measurements was performed. The accidental count rate was also corrected for using the relation [13],

$$\Gamma_{\text{accidental}} \approx \tau \Gamma_0 \Gamma_1, \tag{5}$$

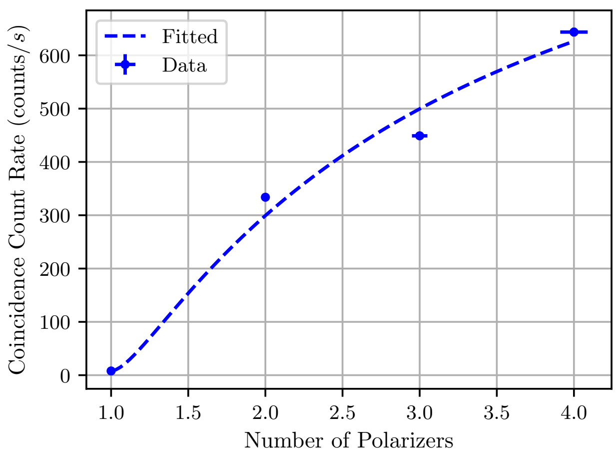

FIG. 2. Coincidence count rate as a function of number of polarizers. Fit performed to Equation 6.

where τ is the coincidence window (here, 40 ns [10]) and Γ_0 and Γ_1 are the single count rates in arm 0 and 1, respectively. This accidental count rate, along with its calculated error through Poisson statistics, was subtracted for each datum. The coincidence count rate as it relates to the number of polarizers is shown in Fig. 2.

These data were fit to a modification of Equation 4,

$$\Gamma(n) = A\cos^{2n}\left(\frac{\pi}{2n}\right) + C, \tag{6}$$

with A and C free fit parameters. The model showed a fair fit to $A = 1250(34)$ counts/s and $C = 8(4)$ counts/s, with $R^2 = 0.9810$ and reduced $\chi^2 = 21.2383$. Such a high reduced χ^2 indicates that we are underestimating the error in our model. For statistical comparison, a linear fit was also performed. The fitted slope was found to be $224(39)\,\mathrm{s}^{-1}$.

For the sake of comparison, the coincidence count rate was also measured for all polarizers set parallel to $|H\rangle$ to simulate the $n \to \infty$ case. This count rate was found to be 1159(11) counts/s in this case. Similarly, the count rate for all polarizers set parallel to $|V\rangle$ was measured to be 14.1(12) counts/s. This gives a measurement of the number of dark counts and counts due to nonideal polarizer transmission.

Conclusion—With one polarizer stage, the angle between the input and output was 90° and shows no transmission, as expected. As the number of polarizers increases and the angle difference decreases, we see a monotonically increasing trend characteristic of the expectations of the QZE. The QZE-free case predicts a zero slope in case of a linear fit, as increasing n would not increase the number of transmitted photons. Statistical analysis of the slope compared to the null reveals a violation of the null hypothesis by $5.80\,\sigma$. Thus, we observe evidence of the QZE.

The fit to the Malus's law prediction is fairly strong, with low statistical confidence arising due to the relatively low number of data points. Also, the limit of the fit function (Equation 6) for $n \to \infty$ is given by $\Gamma \to A+C$, which in this case is 1173(77) counts/s. Comparing to the measured estimated value $n \to \infty$ value of 1159(11) counts/s, we see excellent agreement to $0.18\,\sigma$. The parameter C is approximately equal to the number of dark counts, which agrees to $1.08\,\sigma$; the number of dark counts is fairly underestimated by the fit model.

The accuracy of our measurements are likely affected by imperfect laser alignment, lossy mirrors and polarizers, and nonideal optical fibers. These all decrease the fraction of counts measured in the apparatus compared to theory. This would cause the number of absolute counts to decrease more when the count rate is higher, which is similar to the observed behavior. The lack of physical space to install more polarizers also prevented more data points from being collected. Nonetheless, a strong upwards correlation is observed between coincidence count rate and polarizer stages; this is evidence of the QZE.

Summary—Using coincident photons generated by SPDC, we have demonstrated the presence of the QZE. Increasing the number of polarizers between input and output couplers offset at 90° showed fair agreement to the expected behavior from Malus's law to $R^2 = 98.1\,\%$ and very strong agreement to an upward trend of $5.80\,\sigma$. Thus, we have shown that by placing numerous polarizers with small-angle increments, one can effectively change the polarization of photons without significant loss.

[1] B. Misra and E. C. G. Sudarshan, The Zeno's paradox in quantum theory, J. Math. Phys. **18**, 756 (1977).

[2] A. Peres, Zeno paradox in quantum theory, American Journal of Physics **48**, 931 (1980).

[3] P. Kwiat, H. Weinfurter, T. Herzog, A. Zeilinger, and M. Kasevich, Experimental Realization of Interaction-free Measurements, Annals of the New York Academy of Sciences **755**, 383 (1995).

[4] P. Kwiat, H. Weinfurter, and A. Zeilinger, Quantum Seeing in the Dark, Scientific American **275**, 72 (1996), 24993449.

[5] D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*, 3rd ed. (Cambridge University Press, Cambridge, 2018).

[6] E. Hecht, *Optics*, 5th ed. (Pearson Education, Inc, Boston, 2017).

[7] P. Schuttevaar, Malus' Law Derived from Deterministic Particle Behavior, Journal of High Energy Physics, Gravitation and Cosmology **10**, 958 (2024).

[8] D. N. Klyško and Û. E. Sviridov, *Photons and Nonlinear Optics*, rev. and enlarged ed. (Gordon and Breach, New York London Paris [etc.], 1988).

[9] H. A. Haus, *Waves and Fields in Optoelectronics*, Prentice-Hall Series in Solid State Physical Electronics (Prentice-Hall, Englewood Cliffs, NJ, 1984).

[10] qutools GmbH, quED Datasheet (2022).
[11] qutools GmbH, quED Manual (2021).
[12] qutools GmbH, quCR Manual Version 3.2 (2019).
[13] E. Schroedinger, Rate of n-fold Accidental Coincidences, Nature **153**, 592 (1944).